🎯 Rensto Website AI Agent - Master Plan

Date: November 14, 2025
Status: 📋 Planning Phase
Purpose: Adapt Dima's WhatsApp agent architecture for Rensto website chat agent

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📊 CURRENT WEBSITE STATE

✅ What's Live & Working

Architecture:

- ✅ **Hosting**: Vercel (Next.js) - `rensto.com` is 100% on Vercel
- ✅ **Source**: `apps/web/rensto-site/`
- ✅ **DNS**: Points to Vercel (`cname.vercel-dns.com`)
- ✅ **SSL**: HTTPS working (Vercel managed)
- ✅ **Deployment**: Auto-deploy on git push

Core Pages (All HTTP 200 OK):

Page	URL	Status	Content Quality
Homepage	`/`	✅ Live	Excellent - Shows all 4 service types
Marketplace	`/marketplace`	✅ Live	⚠️ API rate limit issue (shows "Loading...")
Subscriptions	`/subscriptions`	✅ Live	Excellent - Full pricing, features
Custom Solutions	`/custom`	✅ Live	⚠️ Voice UI exists but incomplete
Ready Solutions	`/solutions`	✅ Live	Excellent - Industry niches

Existing Components:

- ✅ `VoiceAIConsultation.tsx` - Voice consultation component (incomplete)
- ✅ `IntelligentOnboardingAgent.tsx` - Onboarding agent component
- ✅ Support page has "Live Chat" button (no implementation)
- ✅ Contact form exists

🚨 CRITICAL GAPS IDENTIFIED

1. No Chat Widget Implementation 🚨

- **Support page** has "Start Chat" button but **NO actual chat widget**
- **No real-time chat** functionality exists
- **No AI agent** integrated into website

2. Voice AI Consultation Incomplete 🚨

- Custom Solutions page has voice UI but **NO backend**
- No n8n workflow connected
- No actual voice consultation working

3. Marketplace API Issues ⚠️

- Airtable rate limiting (5 req/sec)
- Should migrate to Boost.space API
- Currently shows "Loading workflows..." indefinitely

4. Missing Typeform Integrations ⚠️

- Only 1 of 5 Typeforms created
- Custom Solutions page should have Typeform but doesn't

🎯 WHAT WE CAN REUSE FROM DIMA'S WHATSAPP AGENT

✅ Proven Architecture (Working for Dima)

Components We Can Adapt:

1. **RAG System** (Gemini File Search):
   - ✅ Working knowledge base search
   - ✅ File upload and processing
   - ✅ Semantic search with citations
   - **Can reuse**: Search Documents Tool code
2. **AI Agent Architecture**:
   - ✅ LangChain Agent with tools
   - ✅ Memory system (BufferWindow - 10 messages)
   - ✅ Natural Hebrew responses
   - ✅ Tool calling (search_documents)
   - **Can reuse**: Agent node structure, system message patterns
3. **Voice/Text Processing**:
   - ✅ Voice transcription (OpenAI Whisper)
   - ✅ Text-to-speech (ElevenLabs) - optional for website
   - ✅ Message routing (voice vs text)
   - **Can adapt**: For website, we only need text (no voice)
4. **Memory & Context**:
   - ✅ Session-based memory (per user)
   - ✅ Conversation history
   - ✅ Context-aware responses
   - **Can reuse**: Memory node configuration

🏗️ RENSTO WEBSITE AGENT ARCHITECTURE

Proposed System

```
┌──────────────────────────────────────────────────────────┐
│              RENSTO WEBSITE CHAT AGENT                    │
│              (Adapted from Dima's WhatsApp Agent)         │
└──────────────────────────────────────────────────────────┘
                          │
                          │
        ┌─────────────────────────────────┐
        │                                 │
        ↓                                 ↓

┌──────────────────┐              ┌──────────────────┐
│  Website Frontend │              │   n8n Backend     │
│  (Next.js)        │              │   (RackNerd VPS)   │
│                   │              │                   │
│  - Chat Widget    │              │  - Webhook        │
│  - Message UI     │◄─────────────┤  - AI Agent       │
```

```
│  - Input Field    │   HTTP POST  │  - RAG Tool       │
│  - History        │              │  - Memory         │
└──────────────────┘              └──────────────────┘
```

Technology Stack

Frontend (Next.js):

- React chat widget component
- WebSocket or HTTP polling for real-time updates
- Session management (cookies/localStorage)
- Message history UI

Backend (n8n):

- Webhook trigger (receives messages from website)
- LangChain Agent node
- Gemini File Search tool (RAG)
- Memory node (per session)
- HTTP Response node (sends replies back)

Knowledge Base:

- Gemini File Search Store: `fileSearchStores/rensto-knowledge`
- Documents: Service descriptions, pricing, FAQs, technical docs
- Auto-updates from Airtable/Boost.space

📋 IMPLEMENTATION PLAN

Phase 1: Frontend Chat Widget (Week 1)

Tasks:

1. Create chat widget component (`components/chat/RenstoChatWidget.tsx`)
2. Add chat button to all pages (floating button, bottom-right)
3. Implement message UI (bubbles, timestamps)
4. Add input field with send button
5. Implement session management (cookies)
6. Add message history (localStorage or API)
7. Style with Rensto brand colors

Components to Build:

- `RenstoChatWidget.tsx` - Main chat widget
- `ChatMessage.tsx` - Individual message bubble
- `ChatInput.tsx` - Input field with send button
- `ChatHeader.tsx` - Header with agent name/status
- `ChatHistory.tsx` - Message history display

API Endpoint:

- `/api/chat/send` - POST endpoint to send messages
- `/api/chat/history` - GET endpoint to fetch history

Phase 2: n8n Backend Workflow (Week 1-2)

Workflow Name: `INT-WEBSITE-001: Rensto Website Chat Agent`

Node Structure:

```
Webhook Trigger (POST /api/chat/send)
    ↓
Extract Message & Session
    ↓
AI Agent (LangChain)
    ├─→ Simple Memory (session-based, 10 messages)
    ├─→ OpenAI Chat Model (gpt-4o-mini)
    ├─→ Search Documents Tool (Gemini File Search)
    └─→ System Message (Rensto context)
    ↓
Format Response
    ↓
HTTP Response (return to website)
```

System Message (Adapted from Dima's):

```
את Rensto, עוזרת AI ידידותית של פלטפורמת האוטומציה.

**תפקידך:**
- לענות על שאלות על השירותים שלנו (Marketplace, Subscriptions, Ready Solutions, Custom Solutions)
- לעזור למשתמשים לבחור את השירות המתאים להם
- לספק מידע על תמחור, תכונות, ותהליכים
- להפנות למקורות מידע נוספים (Typeforms, דפי שירות)

**איך את מתנהגת:**
- את מדברת בעברית באופן טבעי וידידותי
- את זוכרת את השיחה הקודמת ומתייחסת למה שנאמר קודם
- את לא מתחילה כל הודעה מחדש - את ממשיכה את השיחה
- תגובות קצרות וברורות (פחות מ-150 מילים)

**כשמגיעה שאלה:**
1. קודם כל, את מחפשת במאגר הידע באמצעות search_documents
2. את משתמשת במידע שמצאת כדי לענות
3. אם לא מצאת מידע, את אומרת בכנות שאת לא בטוחה ומפנה למשתמש לטופס יצירת קשר

**חשוב:**
- תמיד תחפשי במאגר לפני שאת עונה
- אם זה המשך של שיחה קודמת, התייחסי למה שנאמר קודם
- אל תחזרי על מה שכבר נאמר בשיחה
- דברי באופן טבעי - לא כמו רובוט, כמו בן אדם
- אם מישהו שואל על שירות ספציפי, תני פרטים מדויקים (תמחור, תכונות, תהליך)
```

Knowledge Base Setup:

1. Create Gemini File Search Store: `fileSearchStores/rensto-knowledge`
2. Upload documents:
   - Service descriptions (Marketplace, Subscriptions, Ready Solutions, Custom)
   - Pricing information
   - FAQ documents
   - Technical documentation
   - Process flows
3. Auto-sync from Airtable/Boost.space (future enhancement)

Phase 3: Integration & Testing (Week 2)

Tasks:

1. Connect frontend to n8n webhook
2. Test message flow (send → n8n → response)
3. Test memory (conversation context)
4. Test RAG (knowledge base search)
5. Test error handling
6. Mobile responsiveness testing
7. Performance optimization

Testing Checklist:

- [ ] Chat widget appears on all pages
- [ ] Messages send successfully
- [ ] Responses received correctly
- [ ] Memory works (remembers previous messages)
- [ ] RAG search works (finds relevant info)
- [ ] Error handling (network errors, API failures)
- [ ] Mobile responsive
- [ ] Session persistence (reload page, history preserved)
- [ ] Multiple concurrent users (session isolation)

Phase 4: Knowledge Base Population (Week 2-3)

Documents to Add:

1. **Service Descriptions**:
   - Marketplace: What it is, how it works, pricing
   - Subscriptions: Lead generation service, pricing tiers
   - Ready Solutions: Industry packages, what's included
   - Custom Solutions: Consultation process, pricing, timeline
2. **FAQ Documents**:
   - Common questions about each service
   - Technical questions
   - Pricing questions
   - Process questions
3. **Process Flows**:
   - How to purchase Marketplace template
   - How to start subscription
   - How to book Custom Solutions consultation
   - How to purchase Ready Solutions package
4. **Technical Docs**:
   - n8n workflow installation
   - API documentation
   - Integration guides

🎨 UI/UX DESIGN

Chat Widget Design

Position: Fixed bottom-right corner (desktop), full-screen overlay (mobile)

Components:

- **Chat Button**: Floating button with Rensto logo/icon
- **Chat Window**: Expandable panel (400px wide, 600px tall)
- **Header**: "Chat with Rensto AI" + close button
- **Messages**: Bubbles (user: right, agent: left)
- **Input**: Text field + send button (bottom)
- **Typing Indicator**: Shows when agent is thinking

Branding:

- Colors: Rensto brand colors (#fe3d51, #bf5700, #1eaef7, #5ffbfd)
- Fonts: Rensto brand fonts
- Icons: Lucide React icons (consistent with site)

States:

- **Closed**: Just floating button
- **Open**: Full chat window
- **Minimized**: Small header bar (can reopen)
- **Loading**: Typing indicator
- **Error**: Error message with retry button

🔄 DIFFERENCES FROM WHATSAPP AGENT

What We DON'T Need:

- ❌ Voice transcription (Whisper) - website is text-only
- ❌ Text-to-speech (ElevenLabs) - website is text-only
- ❌ WAHA integration - we use HTTP webhooks
- ❌ Voice message routing - only text messages

What We DO Need:

- ✅ HTTP webhook (instead of WAHA trigger)
- ✅ Session management (cookies instead of phone number)
- ✅ Chat widget UI (instead of WhatsApp)
- ✅ Real-time updates (WebSocket or polling)

What We CAN Reuse:

- ✅ AI Agent architecture (LangChain)
- ✅ RAG system (Gemini File Search)
- ✅ Memory system (BufferWindow)
- ✅ System message patterns
- ✅ Tool calling (search_documents)
- ✅ Natural language responses

📊 CURRENT WEBSITE PRIORITIES

Immediate (This Week):

1. ✅ Fix Marketplace API (migrate to Boost.space)
2. ✅ Add Typeform to Custom Solutions page
3. 🔄 Build chat widget frontend
4. 🔄 Create n8n workflow for chat agent

Short Term (Next 2 Weeks):

1. Populate knowledge base
2. Test and refine chat agent
3. Add chat widget to all pages

4. Monitor and optimize

Medium Term (Next Month):

1. Add analytics (chat usage, common questions)
2. Improve knowledge base (auto-sync from Airtable)
3. Add chat history in customer portal
4. A/B test different system messages

🚀 NEXT STEPS

Step 1: Create Chat Widget Component

- Location: `apps/web/rensto-site/src/components/chat/RenstoChatWidget.tsx`
- Features: Basic UI, message display, input field
- Integration: Add to layout (appears on all pages)

Step 2: Create API Endpoint

- Location: `apps/web/rensto-site/src/app/api/chat/send/route.ts`
- Purpose: Receive messages from frontend, forward to n8n
- Response: Return agent reply

Step 3: Create n8n Workflow

- Name: `INT-WEBSITE-001: Rensto Website Chat Agent`
- Based on: Dima's WhatsApp agent (`86WHKNpj09tV9j1d`)
- Adaptations: Remove voice nodes, add HTTP webhook

Step 4: Setup Knowledge Base

- Create Gemini File Search Store
- Upload initial documents
- Test search functionality

Step 5: Test & Deploy

- Test end-to-end flow
- Fix any issues
- Deploy to production

📝 FILES TO CREATE/MODIFY

New Files:

1. `apps/web/rensto-site/src/components/chat/RenstoChatWidget.tsx`
2. `apps/web/rensto-site/src/components/chat/ChatMessage.tsx`
3. `apps/web/rensto-site/src/components/chat/ChatInput.tsx`
4. `apps/web/rensto-site/src/components/chat/ChatHeader.tsx`
5. `apps/web/rensto-site/src/app/api/chat/send/route.ts`
6. `apps/web/rensto-site/src/app/api/chat/history/route.ts`
7. `live-systems/n8n-system/workflows/INT-WEBSITE-001.json` (workflow export)

Modify Files:

1. `apps/web/rensto-site/src/app/layout.tsx` - Add chat widget to layout
2. `apps/web/rensto-site/src/app/custom/page.tsx` - Add Typeform integration
3. `apps/web/rensto-site/src/app/api/marketplace/workflows/route.ts` - Fix API (use Boost.space)

🎯 SUCCESS METRICS

Technical Metrics:

- Chat widget loads in < 1 second
- Message response time < 3 seconds
- 99% uptime
- Zero critical errors

Business Metrics:

- Chat engagement rate (users who open chat)
- Questions answered correctly (RAG accuracy)
- Conversion rate (chat → purchase)
- User satisfaction (if we add feedback)

📚 REFERENCES

Dima's WhatsApp Agent:

- Workflow ID: `86WHKNpj09tV9j1d`
- Location: Rensto VPS n8n instance
- Components: WAHA Trigger, AI Agent, RAG Tool, Memory

Documentation:

- `/dima/DONNA_AI_CONVERSATION_SUMMARY.md` - Technical details
- `/dima/DONNA_AI_PROPOSAL.md` - Business context
- `/docs/infrastructure/WEBSITE_CURRENT_STATUS.md` - Website state

Status: 📋 **PLAN READY FOR EXECUTION**
Estimated Time: 2-3 weeks for full implementation
Priority: High (improves user experience, reduces support burden)